A COMPANION TO THE AI ENGINEERING OPERATING MODEL PLAYBOOK

Twelve decisions that decide whether AI engineering pays

They run from platform architecture to organisation design to the business case. Most of them get made early, quietly, and by default.

● Published ● Reasoned ● First-hand ● Judgement

The dots above say what kind of claim each number is. The workbook behind the case is open, illustrative and directional rather than a full P&L, and the field notes are mine with names removed. Competent tool choice and implementation are assumed throughout and are not covered here. August 2026.

Adoption is nearly universal. Measured gain is not.

90%
of developers use AI at work

7.76%
throughput gain across 400 companies

39%
of companies report any EBIT impact at all

Well-chosen, well-implemented tools still do not change a business on their own. The change comes from cross-discipline decisions, which often have no single owner.

Here are twelve worth making on purpose.

PUBLISHED DORA 2025; DX, Nov 2024 to Feb 2026; McKinsey 2025

The 7.76% is pull-request throughput over the fifteen months to February 2026. Most organisations land between 5% and 15%.

The twelve, in the order they arrive.

FRAME

01 Do you measure before you start?

02 Faster engineers, or faster delivery?

BUILD

03 One paved route, or everyone finds their own way?

04 Does the agent see only code, or the thinking behind it?

05 Who decides the output is good enough, and when?

RUN

06 What can you see, cap and stop?

07 Train people to prompt, or to supervise?

08 What do you centralise, and what stays with the teams?

09 Who goes first, and on what?

PROVE

10 Where do freed hours go, and who signs?

11 What do you promise for year one?

12 What would make you stop?

The playbook follows the build process and this deck the decision sequence, so the chapter order differs. Each page opens with the default; where one reads familiar, start there.

Do you measure before you start?

BY DEFAULT

Run a pilot, measure afterwards, compare the result against how it felt.

BY CHOICE

Capture the baseline before the first licence is activated. It cannot be reconstructed later.

AI value follows a J-curve: delivery dips before it climbs. DORA names that dip the **tuition cost** and warns that leaders who misread it as failure pull the funding. A baseline is what tells them apart.

PUBLISHED DORA, The ROI of AI-assisted Software Development, 2026

The cost is low: the metrics are ones §8 already asks for, and the baseline is each team's pre-onboarding period.

Faster engineers, or faster delivery?

BY DEFAULT

Usage leaderboards and acceptance rates, the numbers the tool supplies.

BY CHOICE

What reaches production, with quality on the same chart.

Across 400 companies, AI usage rose **65%** while pull-request throughput rose **7.76%**. Writing is only the first step. Review, integration and coordination all sit downstream, and the gain has to survive them.

PUBLISHED DX longitudinal study, Nov 2024 to Feb 2026

7.76% counts merged pull requests, not what reached production, and nothing in the count separates rework from new work.

One paved route, or everyone finds their own way?

BY DEFAULT

Buy the licences and let local experiments follow.

BY CHOICE

One route through the plumbing, so the experiments are about the product.

Where platform quality is low, AI adoption has a **negligible effect** on organisational performance. Where it is high, the effect is **strong and positive**.

PUBLISHED DORA 2025, platform engineering capability

The test is whether an engineer under deadline still takes the route. If their own way is quicker, the route is not good enough.

Does the agent see only code, or the thinking behind it?

BY DEFAULT

The repository, and whatever rules files have accumulated around it.

BY CHOICE

The decisions, constraints and ownership the code does not record, assembled deliberately and kept current.

One project in my portfolio planned its source-data and context work at roughly a quarter of the total effort. It ran closer to **60%**, which made it the largest single item in the build.

FIRST-HAND anonymised

A business case without the context work understates the cost.

Who decides the output is good enough, and when?

BY DEFAULT

The reviewer decides, case by case, against their own read of what good looks like.

BY CHOICE

The standards written down, served to the agent before it works, and checked automatically after.

Two-thirds of developers name output that is **almost right, but not quite** as their biggest frustration. Almost right is a judgement about whether something fits this estate, and an agent cannot make it without knowing what the estate has already decided.

PUBLISHED Stack Overflow Developer Survey 2025

The playbook calls this assembly a proposal, not yet proven in production. The entry rung is small: a rules file, an ADR set, and structural checks in CI.

What can you see, cap and stop?

BY DEFAULT

A policy document, intended to work as a control.

BY CHOICE

A gateway every call passes through, with approved models behind it, a spend cap that has an exception path, and a kill-switch someone has tested.

At one organisation, AI use was banned at group level. Every person I spoke to used it anyway. What the ban produced was **usage with no visibility, no telemetry and no control**. Even the money is hard to see: industry-wide, FinOps practitioners rank granular monitoring of AI spend as their top missing tool.

FIRST-HAND anonymised; State of FinOps 2026

A gateway does not stop someone determined to go around it. It reduces the reason to, which a ban cannot. What still goes around it tells you where the route falls short.

Train people to prompt, or to supervise?

BY DEFAULT

Prompt technique. How to ask the tool for a better answer.

BY CHOICE

Supervision. How to review generated work, when to reject it, and what must be verified rather than trusted.

Goldman Sachs' CIO frames these as **management skills**: describe, delegate, supervise. He calls delegating to an agent you cannot supervise a recipe for disaster. Reviewing work you did not write, at volume, is a different skill from writing it.

● **PUBLISHED** Marco Argenti, CIO, Goldman Sachs, July 2025

Prompting shapes how you ask. It does not set what the work should draw on, what it must satisfy, or how you would know it is done.

What do you centralise, and what stays with the teams?

BY DEFAULT

One AI team owns the tools, the standards and the rollout. Who owns the value is left open.

BY CHOICE

The team that builds the route answers for adoption. The teams that deliver own the outcome and book the value.

One team should build the platform. Adoption can be seen from there. What the freed time turned into can only be seen where the work happens.

REASONED follows from §9, where booking requires a named owner

Low adoption points at the platform. High adoption with nothing booked points elsewhere. The split does not end the argument, it makes it answerable.

Who goes first, and on what?

BY DEFAULT

Whoever is keenest goes first, on whatever they are already doing.

BY CHOICE

Representative teams, on work that is frequent, easily checked, and contained if it goes wrong.

A volunteer team on familiar work tells you what happens where everything already favours it. A representative team on checkable work tells you where it is actually useful, under conditions you set.

JUDGEMENT argued in §11 rather than measured

Champions matter, and the strongest ones come from teams that did not ask for it. A win there is harder for a sceptic to dismiss.

Where do freed hours go, and who signs?

BY DEFAULT

The hours go into whatever the teams were already doing, and that becomes the decision.

BY CHOICE

Pick the conversion route before the capacity is freed: throughput, cost-out, debt paydown, or time to market. Engineering proposes, the executive committee chooses, finance signs.

Bain finds teams using AI assistants see gains of **10% to 15%**, and that the time saved is often not redirected toward higher-value work.

PUBLISHED Bain, From Pilots to Payoff, 2025

Bain's evidence is assistant-era, and agents free more time. The pull is familiar from the field: in a project I ran in 2017, time formally protected at 50% was squeezed to 10% by business pressure, and winning it back took effort each time.

What do you promise for year one?

BY DEFAULT

Payback inside the first year, with licences standing in for the cost of the programme.

BY CHOICE

A J-curve with the dip in the budget from the start. Say which year you expect breakeven, and let the gates test it.

The cost is a five-line stack: licences and tokens, the enabling team (where §3's context work lands), the platform, training, and instability. Waves spread it, so year one carries partial coverage against the full build cost.

REASONED follows from §12, where the stack is built bottom-up

The workbook works one example through at 300 engineers. Replace every input with your own before anyone quotes a number from it.

What would make you stop?

BY DEFAULT

The programme has a start date and a budget. The stopping decision gets made reactively, when money or patience runs out.

BY CHOICE

Gates at every phase and every wave, each with numbers to clear. A missed gate gets one of three answers agreed with the CFO in advance: hold, shrink, or kill.

Gartner expects over **40%** of agentic AI projects to be cancelled by end-2027, on escalating costs, unclear business value or inadequate risk controls. All three show up at a gate while the programme is still cheap to change.

PUBLISHED Gartner, June 2025

A programme that cannot stop is not governed, it is funded. The harder half is knowing when not to stop: the classic failure is pulling funding at the bottom of the dip.

One thing sits behind all twelve.

None of them is a decision about the model. Every single one is a decision about the system the model lands in.

DORA reaches the same conclusion from nearly 5,000 technology professionals: AI's primary role is an **amplifier**, magnifying an organisation's existing strengths and weaknesses. These twelve are how you decide which it meets.

A better model raises the ceiling. These twelve decide where under it you land.

PUBLISHED DORA 2025, State of AI-assisted Software Development

DORA's finding cuts both ways: a struggling system gets its struggles amplified. §10's readiness matrix exists to find that out before the licences do.

If you can only ask five.

1. Where is the baseline we captured before the first licence?
2. What happens to the bill if usage triples at renewal?
3. Who owns conversion of freed capacity, by name, and what did they book last quarter?
4. Which year do we expect breakeven, and are we on track for it?
5. What would make us stop?

Asked once, they take a snapshot. Asked before each wave is approved, they work as a gate: a question without a credible answer gets a named owner and a date, and the wave waits.

They are decisions 01, 06, 10, 11 and 12: the questions a CFO or COO can ask without knowing anything about engineering, and the ones a CTO should want asked.

THE FULL ARGUMENT, AND THE WORKBOOK BEHIND IT

The playbook and the open workbook

The full argument behind each of the twelve, and an illustrative business case to rebuild with your own headcount and payroll. The workbook downloads from §12, so the assumptions arrive before the numbers.

yspbob.github.io/AI-Playbook/AI_Engineering_Playbook.html

Every claim is labelled by what kind of claim it is. That distinction is the difference between a business case and a hope.

Written by Yaroslav Pavolotskyi, linkedin.com/in/yaroslavpavolotsky. Market data current as of August 2026 and dated in place, because all of it moves. Appendix A of the playbook carries the July 2026 market snapshot in full.